

Information and Network Security

Study Guide

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1.1 Computer Security Concept	3
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1.1 Computer Security Concept

In daily life we use information for various purposes and use network for communication and exchange information between different parties.

In many cases these information are sensitive so we need to take care that only authorized party can get that information.

For maintaining such privacy we require some mechanism or physical device which ensures that it is safe. Such mechanism or physical devices are known as security system.

Computer Security: The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability, and confidentiality of information system resources.

- This definition of computer security introduces three key objectives that are at the heart of computer security:

1. Confidentiality: It covers two concepts

Data Confidentiality: Assures that private or confidential information is not made available or disclosed to unauthorized individuals.

Privacy: Assures that individuals control or influence what information related to them may be collected and stored and by whom and to whom that information may be disclosed.

2. Integrity: It covers two concepts

Data Integrity: Assures that information and programs are changed authorize manner only

System Integrity: Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.

3. Availability:

Assures that systems work promptly and service is not denied to authorize user.

- **Unconditionally secure algorithm:** An algorithm or an encryption scheme is unconditionally secure if the attacker cannot obtain the corresponding plaintext from cipher text no matter how

much ciphertext is available.

• **Computationally secure algorithm:** An encryption scheme is said to be computationally secure if either of the following criteria is met:

- The cost of breaking the cipher exceeds the value of the encrypted information.
- The time required to break the cipher exceeds the useful lifetime of the information.

• **Threat:** A potential for violation of security, which exists when there is a circumstance, capability,

Action, or event that could breach security and cause harm. That is, a threat is a possible danger that might exploit vulnerability

Next Steps

- Explore various security attacks.
- Learn about OSI Architecture.
- Learn about Network model Security.

References:

1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetw0000stal_m1x5

1.2 OSI Security Architecture

OSI Security

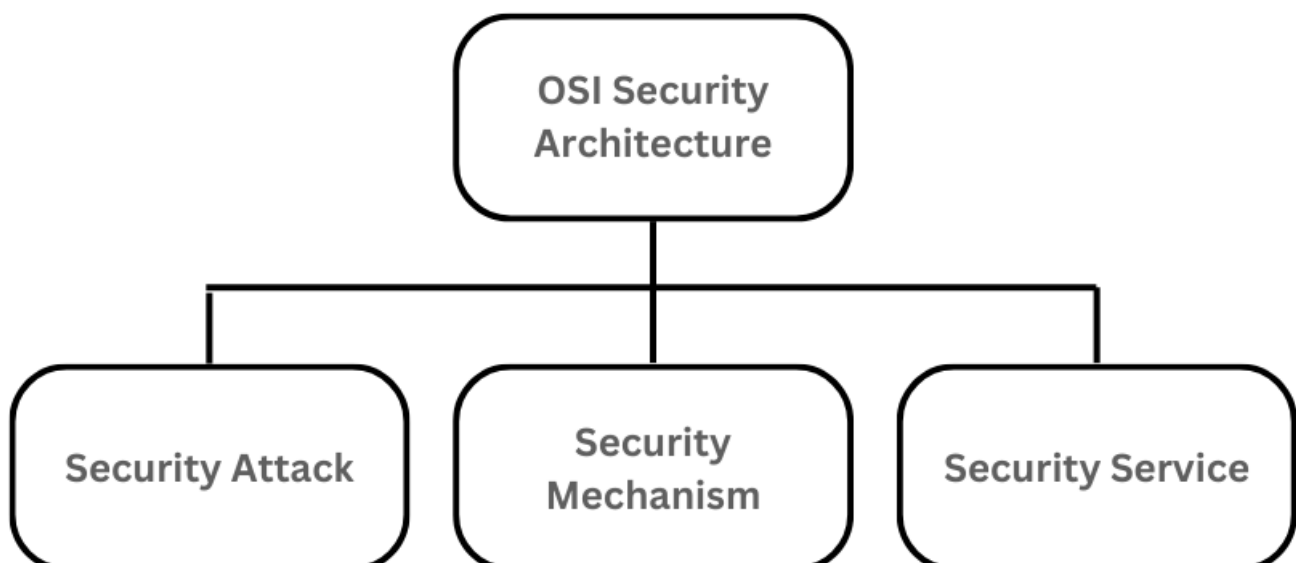
OSI (Open Systems Interconnection) security refers to a set of protocols, standards, and techniques used to ensure the security of data and communications in a network environment based on the OSI model. The International Organization for Standardization (ISO) established this model to provide a conceptual framework for understanding how different networking protocols interact within a layered architecture.

Classification of OSI Security Architecture

The OSI (Open Systems Interconnection) Security Architecture defines a systematic approach to providing security at each layer. It defines security services and security mechanisms that can be used at each of the seven layers of the OSI model to provide security for data transmitted over a network. These security services and mechanisms help to ensure the confidentiality, integrity, and availability of the data. OSI architecture is internationally acceptable as it lays the flow of providing safety in an organization. OSI Security Architecture focuses on these concepts:

1. Security Attack
2. Security mechanism
3. Security Service

OSI Security Architecture is categorized into three broad categories namely Security Attacks, Security mechanisms, and Security Services.



Security Attacks

A security attack is an attempt by a person or entity to gain unauthorized access to disrupt or compromise the security of a system, network, or device. These are defined as the actions that put at risk an organization's safety. They are further classified into 2 sub-categories:

Passive Attack: Attacks in which a third-party intruder tries to access the message/ content/ data being shared by the sender and receiver by keeping a close watch on the transmission or eave-dropping the transmission is called Passive Attacks.

Active Attacks: Active attacks refer to types of attacks that involve the attacker actively disrupting or altering system, network or device activity. Active attacks are typically focused on causing damage or disruption rather than gathering information or intelligence.

Security Mechanism

The mechanism that is built to identify any breach of security or attack on the organization, is called a security mechanism. Security Mechanisms are also responsible for protecting a system, network, or device against unauthorized access, tampering, or other security threats.

Security Services

Security services refer to the different services available for maintaining the security and safety of an organization. They help in preventing any potential risks to security

Next Steps

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References:

1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetw0000stal_m1x5
3. OSI Model : <https://www.geeksforgeeks.org/osi-security-architecture/>

1.3 Security Attacks

Security Attacks

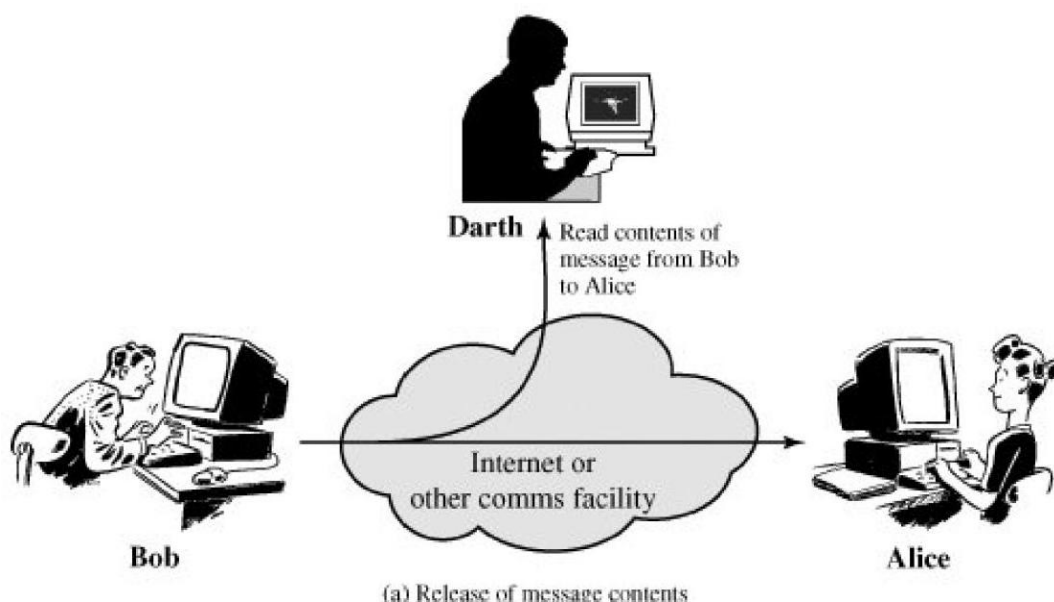
- Security Attacks: An attack is an action that comprises the information or network security.
- There are two types of attacks:
 1. Passive Attack
 2. Active Attack

Passive Attack

- Passive Attack: The attacker only monitors the traffic attacking the confidentiality of the data. It contains release of message contents and traffic analysis (in case of encrypted data).

1. Release of message contents:

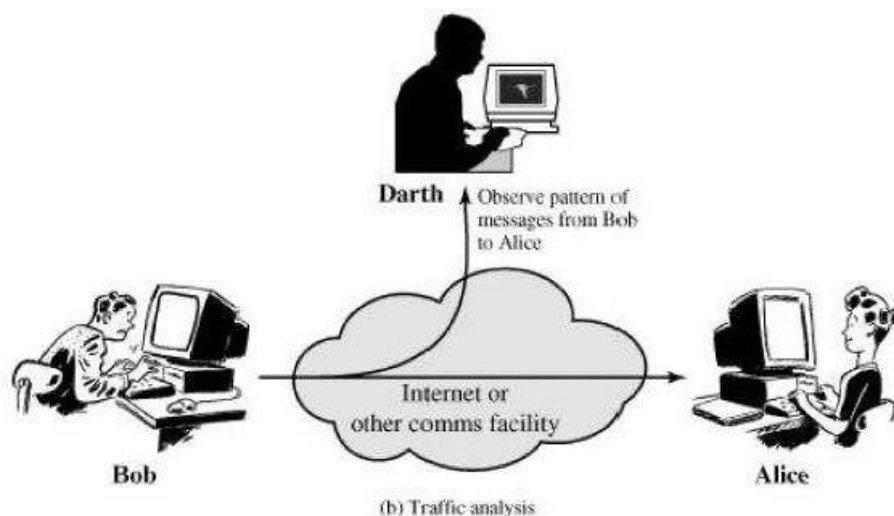
- The release of message contents is easily understood.
- A telephone conversation, an electronic mail message, and a transferred file may contain sensitive or confidential information.
- We would like to prevent an opponent from learning the contents of these transmissions.



2. Traffic analysis:

- A second type of passive attack, traffic analysis.

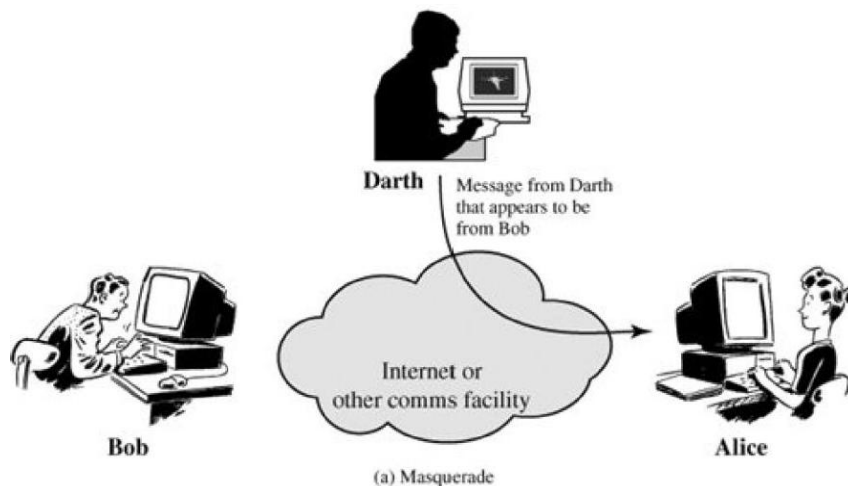
- Suppose that we had a way of masking the contents of messages or other information.
- Even if they captured the message, could not extract the information from the message.
- The common technique for masking contents is encryption.
- If we had encryption protection in place, an opponent might still be able to observe the pattern of these messages.
- The opponent could determine the location and identity of communicating hosts and could observe the frequency and length of messages being exchanged.
- This information might be useful in guessing the nature of the communication that was taking place.
- Passive attacks are very difficult to detect because they do not involve any alteration of the data.
- Typically, the message traffic is send and received in an apparently normal fashion and the sender nor receiver is aware that a third party has read the messages or observed the traffic pattern.



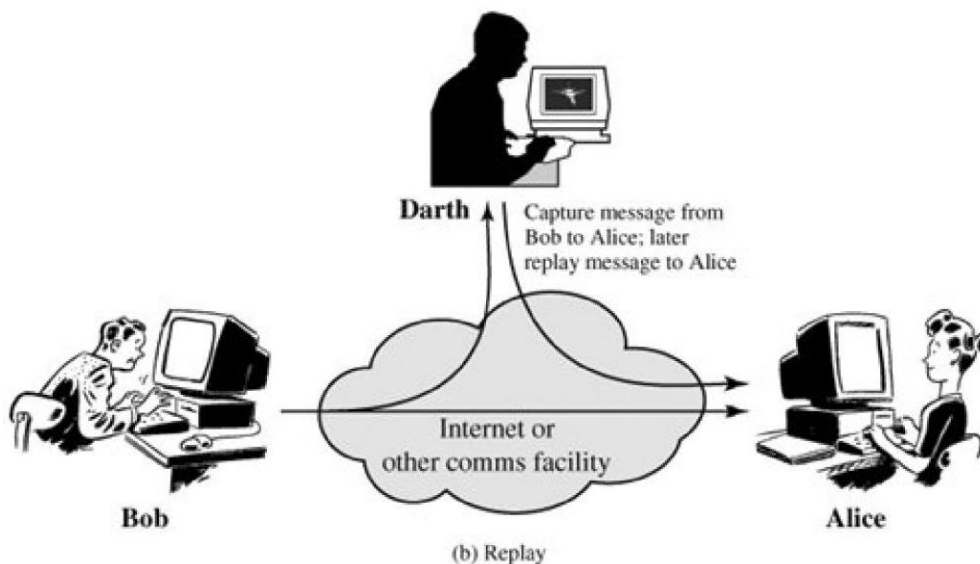
Active attack

- Active attack: Attacker tries to alter transmitted data. It includes masquerade, modification, replay and denial of service.

1. Masquerade: A masquerade takes place when one entity pretends to be a different entity (Figure a). A masquerade attack usually includes one of the other forms of active attack.



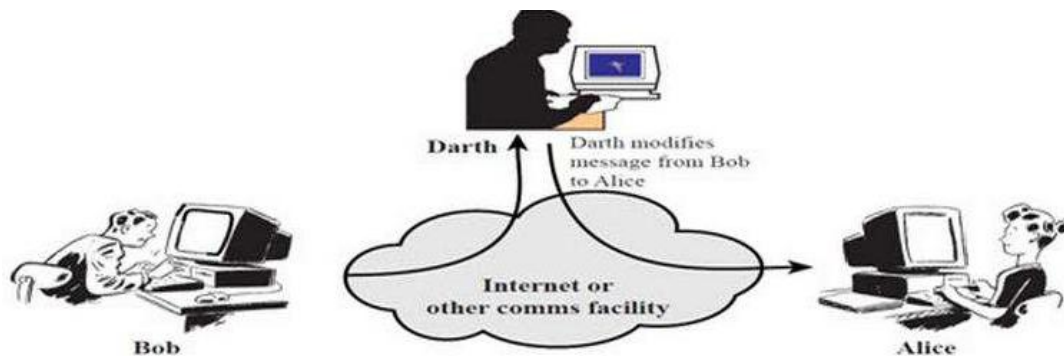
2. Replay: Replay involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.



3. Modification of messages:

- Modification of messages simply means that some portion of a legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect (Figure c).

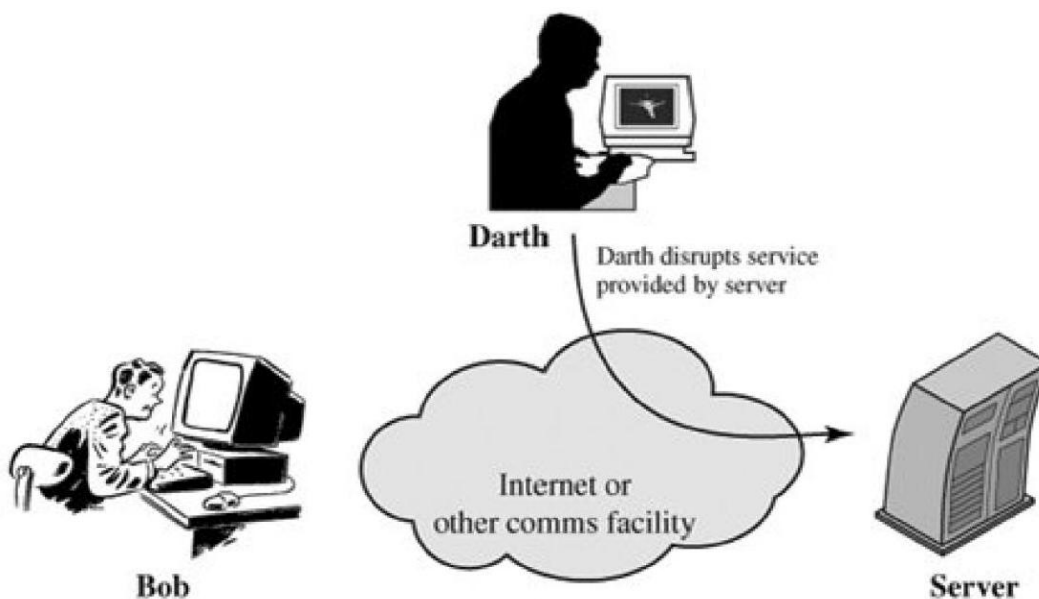
- For example, a message meaning "Allow John Smith to read confidential file accounts" is modified to mean "Allow Fred Brown to read confidential file accounts."



(c) Modification of messages

4. Denial of service:

- The denial of service prevents or inhibits the normal use or management of communications facilities.
- This attack may have a specific target; for example, an entity may suppress all messages directed to a particular destination (e.g., the security audit service).
- Another form of service denial is the disruption of an entire network, either by disabling the network or by overloading it with messages so as to degrade performance.



(d) Denial of service

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1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetw0000stal_m1x5
3. Security Active Attacks :
https://www.tutorialspoint.com/cryptography/attacks_on_cryptosystems.htm

1.4 Security Services

A security service is a processing or communicating service that can prevent or detect the above- mentioned attacks. Various security services are:

- **Authentication:** the recipient should be able to identify the sender, and verify that the sender, who claims to be the sender, actually did send the message.
- **Data Confidentiality:** An attacker should not be able to read the transmitted data or extract data in case of encrypted data. In short, confidentiality is the protection of transmitted data from passive attacks.
- **Data Integrity:** Make sure that the message received was exactly the message the sender sent.
- **Nonrepudiation:** The sender should not be able to deny sending the should not be able to deny receiving the message.

Next Steps

- Learn about Network model Security.

References:

1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetwork0000stal_m1x5
3. Security Services : <https://www.sanfoundry.com/security-services-in-cryptography/>

1.5 Security Mechanism

Security Mechanism Types

1. Encipherment:

This security technique helps make data confidential by concealing and covering it. It is accomplished by using algorithms or mathematical computations to reassemble information into an unintelligible format. Two well-known methods called encryption and cryptography are used to accomplish this. The Encipherment algorithm determines the level of data encryption.

2. Access Control:

This feature prevents unauthorized access to the data you send. There are several ways to accomplish this, like employing firewalls, applying passwords, or simply adding PINs to data.

3. Notarization:

This security measure entails communicating with a reliable third party. It serves as a mediator between the sender and the recipient, minimizing the likelihood of conflict. Requests made by the sender to the recipient are recorded by this mediator in case they are later rejected.

4. Data Integrity:

This protection technique adds value to data that is generated by the data itself. It is comparable to transmitting a packet of data that is known to both the sender and the recipient and is verified both before and after the data is received. Data integrity is preserved when the added packet or data is verified to be identical during transmission and reception.

5. Authentication Exchange:

Identity recognition in communication is addressed by the authentication exchange security mechanism. This is accomplished at the TCP/IP layer by using a two-way handshaking method to verify whether or not data is transmitted.

6. Bit stuffing:

This security technique involves inserting extra bits into transmitted data. Even or odd parity is used to do this, which aids in data verification at the receiving end.

7. Digital Signature:

By including digital information that is invisible to the naked eye, this security measure is accomplished. This type of electronic signature is inserted by the sender and electronically verified by the recipient. Although the sender's identity must be disclosed, this approach is utilized to maintain material that is not more secret.

Next Steps

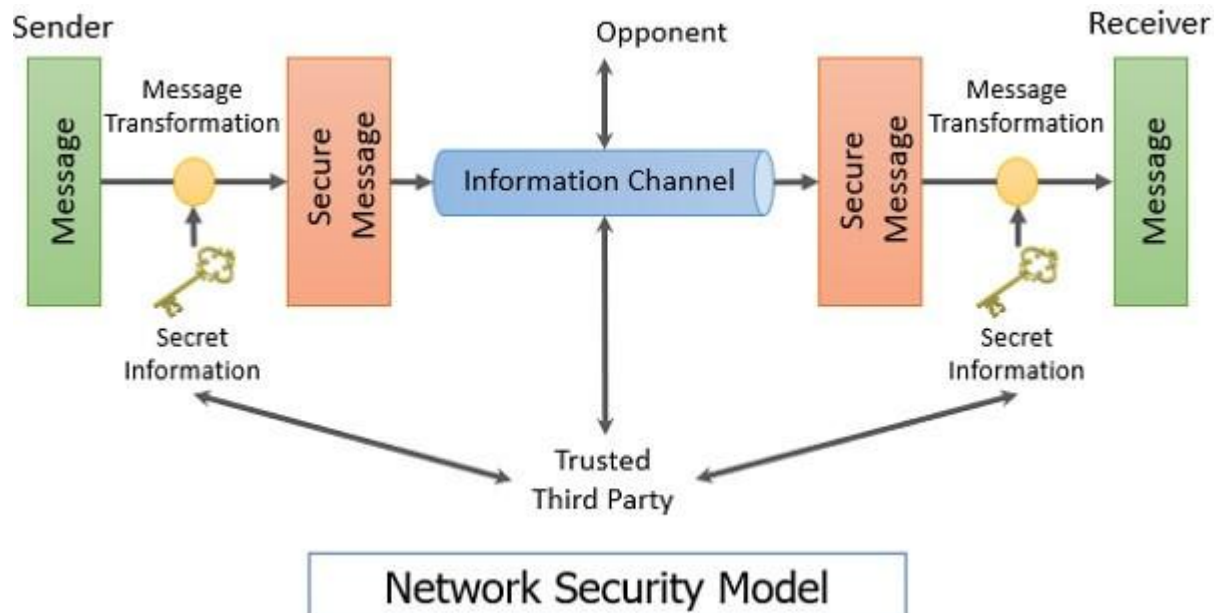
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References:

1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetw0000stal_m1x5
3. Security Mechanism : <https://www.geeksforgeeks.org/types-of-security-mechanism/>

1.6 Model for Network Security

A symmetric cipher model are broadly contains five parts.



1. Plaintext: This is the original intelligible message Encryption algorithm:
 2. The encryption algorithm performs various substitutions and transformations on the plaintext. It takes in plaintext and key and gives the ciphertext.
 3. Secret key: The key is a value independent of the plaintext and of the algorithm. Different keys will yield different outputs.
 4. Ciphertext: This is the scrambled message produced as output. It depends on the plaintext and the secret key
 5. Decryption algorithm: Runs on the ciphertext and the key to produce the plaintext. This is essentially the encryption algorithm run in reverse.
- Two basic requirements of encryption are:
 1. Encryption algorithm should be strong. An attacker knowing the algorithm and having any number of ciphertext should not be able to decrypt the ciphertext or guess the key.

2. The key shared by the sender and the receiver should be secret.

- Let the plaintext be $X = [X_1, X_2, \dots, X_M]$, key be $K = [K_1, K_2, \dots, K_J]$ and the ciphertext produced be $Y = [Y_1, Y_2, \dots, Y_N]$. Then, we can write $Y = (K, X)$

Here E represents the encryption algorithm and is a function of plaintext X and key K.

The receiver at the other ends decrypts the ciphertext using the key.

$$X = D(K, Y)$$

Here D represents the decryption algorithm and it inverts the transformations of encryption algorithm.

An opponent not having access to X or K may attempt to recover K or X or both.

It is assumed that the opponent knows the encryption (E) and decryption (D) algorithms.

If the opponent is interested in only this particular message, then the focus of the effort is to recover by generating a plaintext estimate $\hat{X}$.

If the opponent is interested in being able to read future messages as well then he will attempt to recover the key by making an estimate $\hat{K}$.

Cryptography

Cryptography: The area of study containing the principles and methods of transforming an intelligible message into one that is unintelligible, and then retransforming that message back to its original form.

Cryptographic systems are characterized along three independent dimensions.

The types of operations used for transforming plaintext to cipher text.

All encryption algorithms are based on two general principles substitution, and transposition. Basic requirement is that no information be lost. Most systems referred to as product system, involves multiple stages of substitutions and transpositions.

The number of keys used. If both sender and receiver use the same key, the system is referred to as symmetric, single-key, secret-key, or conventional encryption. If the sender and receiver use different keys the system is referred to as asymmetric, two-key, or public-key encryption.

The way in which the plaintext is processed. A block cipher process a block at a time and produce an output block for each input block. A stream cipher process the input element continuously, producing output one element at a time, as it goes along.

References:

1. Computer Security : https://en.wikipedia.org/wiki/Computer_security
2. Cryptography and network security : principles and practices
https://archive.org/details/cryptographynetw0000stal_m1x5
3. Network Security Model : <https://binaryterms.com/network-security-model.html>